ESV Project Proposal

Robust Mathematical & Coding Reasoning Through Error Identification, Diagnosis, and Recovery Under Structural Perturbations Through Explicit Strategy Verification (ESV)

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Main Idea

Our goal is to test whether an LLM can tell when its reasoning strategy has become invalid, diagnose why, and recover, and whether Explicit Strategy Verification (ESV) makes recovery more reliable than Direct QA or Chain-of-Thought.

RQ1 Can models recognize their own reasoning failures?

RQ2 Can they diagnose why the original strategy is no longer valid?

RQ3 Can they adapt their strategy and recover?

RQ4 Does ESV beat Direct QA and CoT, across benchmarks?

The 3 × 2 Evaluation Matrix

	Original problem P	**Perturbed problem P′**
Direct QA	Answer only: baseline accuracy on the unmodified problem	Does accuracy drop once the familiar strategy breaks?
CoT	Step-by-step reasoning before the answer	Does explicit reasoning alone add robustness?
ESV	Verify the strategy, then recognize, diagnose and recover	Can the model notice its strategy is invalid and recover?

Our proposed 3 × 2 evaluation matrix (per model, per benchmark)

Literature Review + Gaps (Perturbation Robustness)

JEEBench (Arora et al., 2023): https://arxiv.org/abs/2305.15074

Summary: A benchmark of challenging JEE Advanced problems in physics, chemistry and mathematics. Even the strongest models scored far below top human candidates, with failures in long-horizon algebra and in retrieving the right concepts.

Gap: Scores only the final answer. It cannot tell whether a model noticed its mistake, understood it, or could recover from it.

MATH-PERTURB (Huang et al., 2025): https://arxiv.org/abs/2502.06453

Summary: Applies simple and hard perturbations to competition math problems. Hard perturbations change the problem so the original solution no longer applies, and model accuracy drops sharply: models blindly reuse familiar solution strategies.

Gap: Diagnoses the failure mode but proposes no remedy. It does not measure whether models can recognize, diagnose or recover from a broken strategy.

Literature Review + Gaps (Self-Correction)

LLMs Cannot Self-Correct Reasoning Yet (Huang et al., 2024): https://arxiv.org/abs/2310.01798

Summary: Shows that intrinsic self-correction (asking a model to review its own answer, without external feedback) does not reliably improve reasoning and can even make it worse; earlier reported gains relied on oracle labels.

Gap: Tests generic "review your answer" prompting. It does not test structured, strategy-level verification, nor measure recognition, diagnosis and recovery as separate stages.

The gap our project fills

- Recognition, diagnosis and recovery have so far been studied in isolation, never as one lifecycle.
- No benchmark pairs each problem with a verified strategy-breaking perturbation and scores every stage.
- ESV replaces generic self-review with explicit checks of the conditions a strategy depends on.

Our Plan for Building Strategy-Breaking Perturbations

1

Step 1: Generate

Model A (DeepSeek V4 Flash) reads P and its answer, names the strategy S that solves it, and changes one structural condition so S breaks. It outputs P′, the changed condition and an alternative strategy S′.

2

Step 2: Verify blind

Models B (Qwen 3.5 35B) and C (GPT-OSS-120B) solve P′ without seeing P or Model A's answer, then apply the original strategy S to P′. Both must show that S now fails.

3

Step 3: Validate & freeze

B and C must agree on the answer to P′ with confidence ≥ 0.6. Accepted pairs (P, A, S) → (P′, A′, S′) are frozen and shown identically to every method and model.

Pilot yield

16 / 20

JEEBench math problems accepted as frozen pairs (80%)

6 MCQ · 5 multi-correct MCQ · 5 integer

≤ 3 generation attempts per problem; rejected candidates get the verifiers' feedback

3 model families (DeepSeek, Qwen, OpenAI): the generator never grades its own perturbation

What Is a Strategy-Breaking Perturbation?

A perturbation modifies one structure, relationship or constraint of a problem, so that the strategy that worked for the original no longer applies, while the new problem stays well-defined and solvable.

Original P (JEE Adv. 2016)

$-\pi/6 < \theta < -\pi/12$. $\alpha_1 > \beta_1$ are roots of $x^2 - 2x\sec\theta + 1 = 0$, $\alpha_2 > \beta_2$ of $x^2 + 2x\tan\theta - 1 = 0$. Find $\alpha_1 + \beta_2$.

Strategy: $\tan\theta < 0$, so $\alpha_1 = \sec\theta - \tan\theta$, $\beta_2 = -\tan\theta - \sec\theta$

Answer: $\alpha_1 + \beta_2 = -2\tan\theta$ **(C)**

→

Perturbed P'

Same equations, but now $\pi/12 < \theta < \pi/6$.

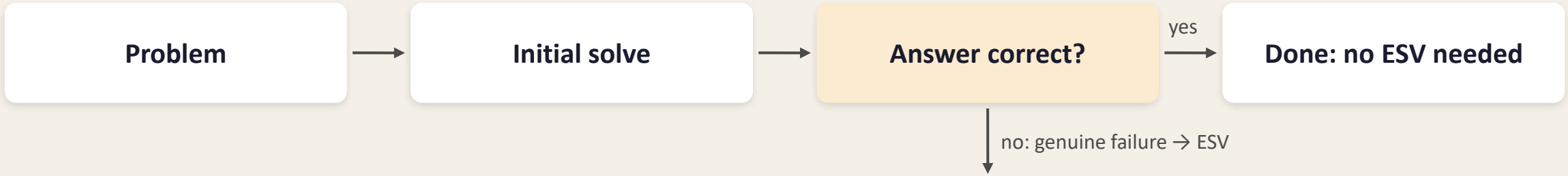
What breaks: $\tan\theta$ is now positive, so the larger root changes: $\alpha_1 = \sec\theta + \tan\theta$

Reusing the old strategy: still answers (C), which is wrong

Correct answer: $\alpha_1 + \beta_2 = 0$ **(D)**

TLDR: the numbers barely change, but a sign condition the old strategy silently relied on is flipped. A model that pattern-matches gets it wrong.

ESV Pipeline Structure



1 Recognition

Blind: re-solve independently and check every condition the strategy relies on. Did it fail?

2 Diagnosis

Find the failed step and the problem condition it conflicts with.

3 Alternative strategy

Propose 3 different methods, verify each one's preconditions, keep the best plus a backup.

4 Recovery

Solve afresh with the verified strategy, without seeing the failed reasoning.

5 Final answer

Cross-check with the backup strategy; adjudicate if the two answers disagree.

Evaluation: answers are scored against the frozen gold answers; an independent judge (GPT-OSS-120B) grades diagnosis and strategy using the gold answer and the perturbation annotation. The solver never sees either.

Pilot Experiment

For our pilot we ran the full ESV pipeline on our 16 frozen JEEBench math pairs: 32 instances. The solver answers each instance once; every wrong answer is sent through all five ESV stages, so every stage is measured on every failure. We built the pipeline in three iterations and report the final one.

1. **Dataset:** 16 original/perturbed pairs (32 instances) from JEE Advanced 2016
2. **Solver:** Qwen 2.5 7B Instruct, chosen by screening: DeepSeek V4 Flash solved 97%, leaving nothing to recover
3. **Judge:** GPT-OSS-120B, a different model family; grades diagnosis and alternative strategy
4. **Scoring:** Deterministic: option letters and numbers normalized, then compared with the gold answer
5. **Blindness:** Recognition is blind; later stages know the first answer was wrong, never the correct one
6. **Iterations:** v2: one gated pass · v3: five stages on every failure · v4: strengthened verification
7. **Baselines:** Direct QA and CoT on the same 16 pairs, run by Ritvij in a separate pipeline

Mini Experiment Figure

1 The test cases

16 frozen pairs
→ 32 instances
P′ generated by Model A,
verified blind by B and C

2 Initial solve

Qwen 2.5 7B answers once
13 correct (40.6%) → done
19 wrong → ESV

3 ESV stages (19 failures)

Recognized: 15
Correct diagnosis: 3
Valid strategy: 7
Recovered: 5

4 Answer & grade

Deterministic scoring
+ judge for stages 2-3
Final: 18 / 32 correct
(56.2%)

Scope & limitations

BY DESIGN: ESV runs only on failures, so after blind recognition the model knows its first answer was wrong (never the correct one).

JUDGE-GRADED: diagnosis and strategy correctness come from an LLM judge given the gold answer; spot-checked, not human-verified.

HONEST LIMITATION: 19 failures on a 16-pair development set, and the Direct QA / CoT baselines come from a separate pipeline whose settings are not yet aligned with ESV.

Results

CoT led in the baseline runs: 68.8% overall vs 56.2% for ESV and 28.1% for Direct QA, so RQ4 is not yet supported.

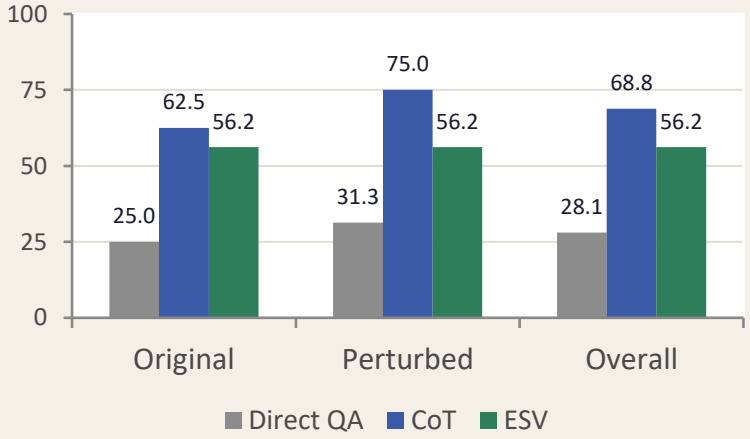
ESV recovered 5 of 19 failures, lifting its own accuracy from 40.6% to 56.2%.

Recognition is ESV's strongest signal: 79% of failures caught without being told.

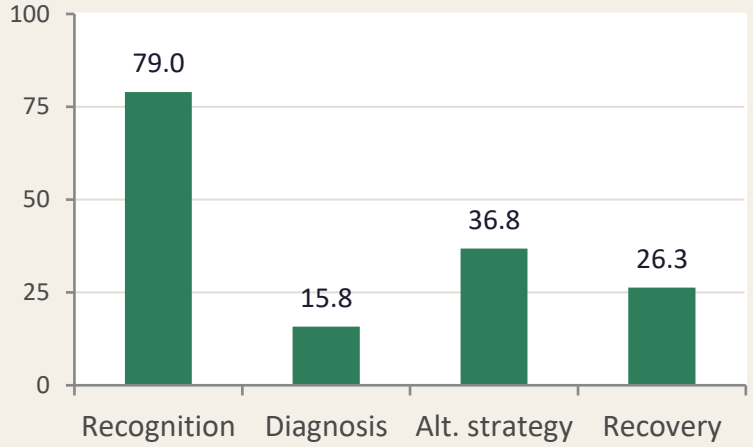
No accuracy drop on perturbed problems for any method, so these perturbations did not break the strategies.

Diagnosis (16%) is the bottleneck for a 7B model.

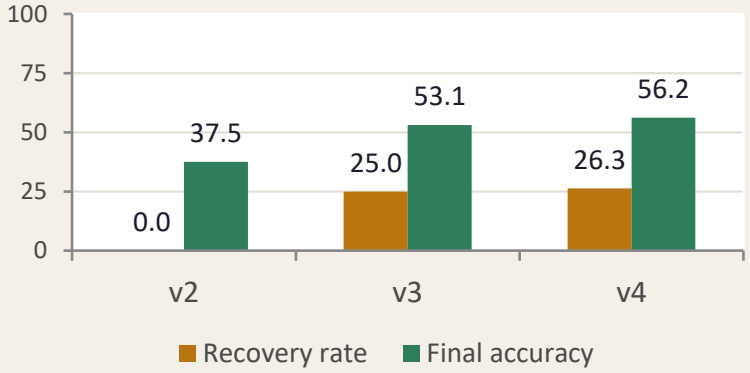
A Accuracy by method (%)



B ESV stage accuracy, 19 failures (%)



C Pipeline iterations (%)



+15.6 pp

ESV gain over its own initial solve: 13 → 18 of 32 problems correct

ESV: Qwen 2.5 7B Instruct, judge GPT-OSS-120B. Direct QA / CoT: reported from a separate pipeline; settings not yet aligned. 16 JEEBench pairs, development set.